Aarush Ram Anandh

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EXECUTIVE SUMMARY

Machine learning researcher working on diffusion models and controllable optimal transport.
Public preprint, C++/Lean research implementation, and controlled reinforcement learning studies.
Python/PyTorch research across vision, RL and asthma detection at approximately 70% sensitivity.

EDUCATION

Ashoka University — B.Sc. (Hons.), Computer Science and Mathematics **Expected 2028**

RESEARCH EXPERIENCE

Constrained Image-Synthesis Lab, Ashoka University **Sep 2025–Present**
Research Fellow

- Improved latent-diffusion colour fidelity to $\Delta E_{00} < 1$, with approximately **4× better colour accuracy** than evaluated open-source baselines; results reported in the preprint below.
- Developing **Controlled Generalised Sliced Wasserstein (CGSW)**, an optimal-transport framework with control-conditioned probes, projections and slice aggregation; paper submitted to **JMLR**.
- Implementing **C++** methods and machine-verified **Lean** specifications/proofs for CGSW.
- Conducting a **CGSW experiment** on hyperspectral waste images with **224 spectral bands and seven categories**, constructing object-level spectral measures to study class separability.

Koita Centre for Digital Health, Ashoka University **May 2026–Present**
Research Intern (part-time)

- Contributed to a wearable-data pipeline achieving approximately **70% sensitivity (recall)** for **asthma exacerbation detection** in study evaluation.
- Built HR/HRV smoothing, residual and change-point features for **191 patients**; implemented patient-wise 80:20 splits, five seeds and patient-bootstrap confidence intervals.

SELECTED PROJECTS

Reinforcement Learning for Pokemon Battles **Ongoing**
Two-Person RL Coursework Project [GitHub]

- Co-developed a **Gymnasium/poke-env** environment with **763 observation features**, legality masks over 26 actions, potential-based reward shaping, and **MaskablePPO and masked DQN** agents.
- Evaluated PPO across **two variants and five seeds each** at 200,000 steps/run; achieved **99.7–100% wins** against the fixed heuristic (1,020 final battles/run), with convergence and uncertainty analysis.

MEIKURAL: Hindustani Raag Recognition **Ongoing**
Audio Classification / Signal Processing [GitHub]

- Developing a **Python/PyTorch** pipeline for **seven Hindustani raags** using the Saraga Hindustani dataset; implemented resampling, RMS normalisation and overlapping **30-second audio windows**.
- Explored per-window and full-recording tonic alignment against native-frequency **STFT/CNN** representations; trained a native-STFT CNN on **3,471 windows over 30 CUDA epochs**.

APPLIED EXPERIENCE

Avartan Labs **May–Jul 2026**
Product Engineering / Machine Learning Intern

- Doubled rare-class correct detection from **40% to 80%** in an **18-class RT-DETR** sorting pipeline.
- Investigated focal loss, logit adjustment, class-specific thresholds and targeted fine-tuning for rare classes; evaluated detector failure modes through crop-based refinement.

PUBLICATION

Inference-Time Loss-Guided Colour Preservation in Diffusion Sampling (2026).
Public preprint: arXiv:2601.17259; colour-fidelity research described above.

TECHNICAL SKILLS

Programming: Python, C++, C, Bash, Lean. **Frameworks:** PyTorch, Diffusers, TensorFlow.
Methods: Optimal transport, diffusion models, reinforcement learning, computer vision, time-series analysis.